

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			from the equation 552 g of Ag ₂ CO ₃ gives 432 g of Ag (1) 1 g of Ag ₂ CO ₃ gives $\frac{432}{552}$ g of Ag (1) 13.8 g of Ag ₂ CO ₃ $\frac{432}{552} \times 13.8$ g of Ag = 10.8 g (1) ecf possible alternative method moles of Ag ₂ CO ₃ = $\frac{13.8}{276} = 0.05$ mol (1) 2 mol of of Ag ₂ CO ₃ gives 4 mol of Ag therefore 0.10 mol of Ag produced (1) mass of Ag = n × M _r = 0.10 × 108 = 10.8 g (1) ecf possible second alternative method (applying conservation of mass) percentage of Ag in Ag ₂ CO ₃ = $\frac{216}{276} \times 100 = 78.3\%$ (1) mass of Ag on left-hand side = $\frac{78.3}{100} \times 13.8 = 10.8$ g (1) so 10.8 g of Ag formed (1) ecf possible		3		3		3
				Question 9 total	0	5	3	8	3	3